D2.3 Water potential: data-based questions

NAME | DATE | TOTAL _____ / 9

Read the data carefully before you answer. The number of marks shows how many separate points to make. Show your working in calculations.

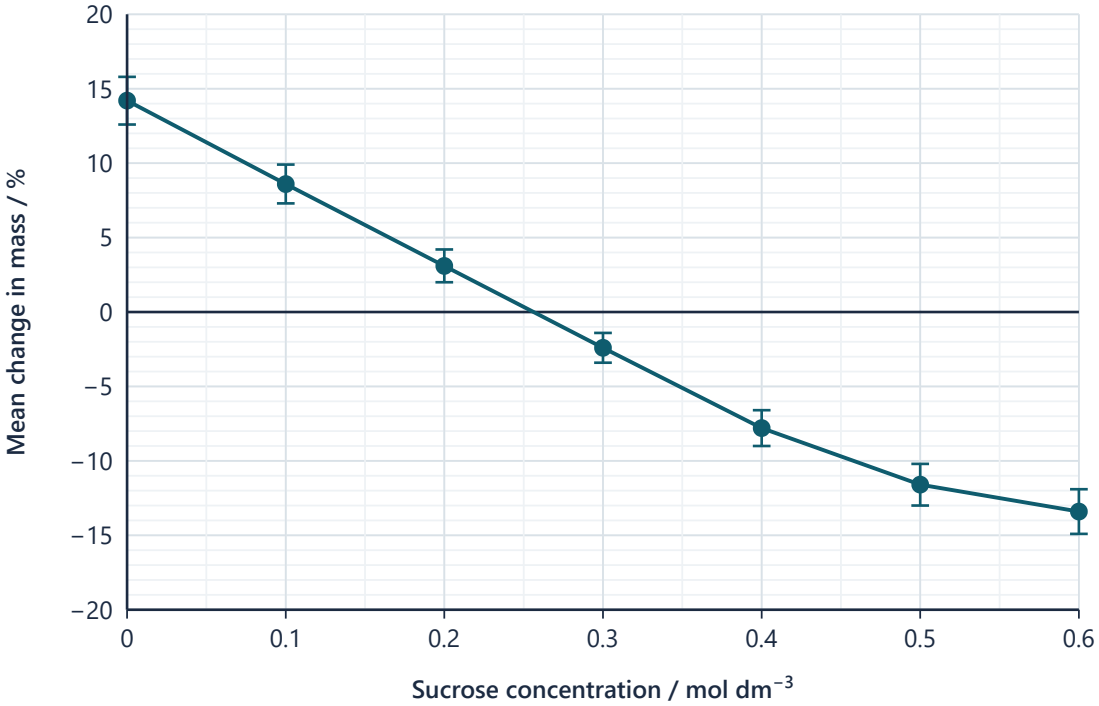
1 Osmosis in potato tissue

9 MARKS



Image: ZooFari, Public domain, Wikimedia Commons.

Cylinders of potato tuber (*Solanum tuberosum*), each about 2.5 g, were blotted, weighed and placed in sucrose solutions at 20 °C. After 24 hours they were blotted and weighed again. Three cylinders were used for each concentration. The graph shows the mean percentage change in mass. Error bars show ± 1 standard deviation.



a ESTIMATE the sucrose concentration that has the same water potential as the potato cells. [1]

b One cylinder in 0.4 mol dm^{-3} sucrose had a mass of 2.48 g at the start and 2.29 g at the end. **Calculate** its percentage change in mass. [1]

c EXPLAIN why the results are given as percentage change in mass, not change in mass. [2]

d EXPLAIN the change in mass of the cylinders in 0.0 mol dm^{-3} sucrose. [2]

e SUGGEST why the mass changes less between 0.5 and 0.6 mol dm^{-3} than between 0.0 and 0.1 mol dm^{-3}. [1]

f Red blood cells are placed in distilled water. PREDICT what happens to them, and **explain** how this differs from the potato cells. [2]